

Part G: Linux Security Research

1. Why are file permissions important?

File permissions help control who can read, modify, or execute files in a Linux system. They protect sensitive information from unauthorized access. Permissions also prevent accidental changes to important files. Proper permissions improve system security and data privacy. They are an essential part of Linux administration.

2. What happens if sensitive files are given 777 permissions?

A file with 777 permissions can be read, modified, and executed by anyone on the system. If sensitive files have these permissions, unauthorized users can view or change their contents. This can lead to data theft, file corruption, or security breaches. Therefore, 777 permissions should be avoided for important files. It is considered a major security risk.

3. What is the Principle of Least Privilege?

The Principle of Least Privilege means that users should only be given the permissions they need to perform their tasks. They should not receive additional access rights that are unnecessary. This reduces the chances of accidental mistakes

and security incidents. It also limits the damage that can occur if an account is compromised. Organizations use this principle to improve security.

4. Why do organizations restrict user access?

Organizations restrict user access to protect important data and systems. Not every employee needs access to all files and resources. Limiting access reduces the risk of unauthorized changes, data leaks, and security threats. It also helps maintain accountability by ensuring users only work with the resources they need. This improves overall system security and management.

Conclusion

Linux file permissions play a vital role in protecting data and maintaining system security. By assigning appropriate permissions and following security principles such as Least Privilege, organizations can reduce risks and ensure that sensitive information remains protected.